

Clustering Model Selection in R: Metrics and Hyperparameter Tuning

Two Different Questions

Two complementary techniques for the analyst who has already fit a clustering model and wants to control how it is judged, and how its hyperparameters are chosen. Both keep the exact same workflow — configure, fit, cluster, evaluate — and change only what happens around it.

Custom Evaluation

Which metrics should judge this partition?

`clu_tune`

Which hyperparameter configuration should I keep?

Customizing Evaluation Metrics

Every clustering model in daltoolbox carries default internal and external evaluation metrics from its class constructor. Those defaults can be inspected and replaced when the analyst wants a narrower or different evaluation view.

```
model <- cluster_kmeans(k = 3)
model$eval_internal <- list(
  model$clu_utils$metric_silhouette)
model$eval_external <- list(
  model$clu_utils$metric_entropy)

model <- fit(model, x)
clu <- cluster(model, x)
evaluate(model, clu, ref)
```

- **eval_internal:** metrics computed from the data and the partition alone — e.g. silhouette.
- **eval_external:** metrics computed against a reference partition, when one exists — e.g. entropy.
- **Discoverable:** available metric functions can be listed via `grep("^metric_", names(model$clu_utils), value = TRUE)`.

Reference: MacQueen (1967); Rousseeuw (1987), Silhouettes; Shannon (1948).

Hyperparameter Tuning – clu_tune

Selects hyperparameters for a clustering method by wrapping a base model and searching a configuration range — the same tuning idea already familiar from `cla_tune` and `reg_tune`, applied to unsupervised learning.

```
base_model <- cluster_kmeans(k = 2)
base_model$metric <-
  base_model$clu_utils$metric_silhouette
base_model$selector <-
  base_model$clu_utils$selector_best

model <- clu_tune(base_model,
  ranges = list(k = 2:10))
model <- fit(model, x)
model$k
```

- **Same wrapping idea:** wraps any base clustering model the same way `cla_tune`/`reg_tune` wrap a supervised learner.
- **Didactic simplification:** tunes `k` against a compact internal criterion (silhouette) while still reporting an external score (entropy).
- **Picks the winner:** `model$selector` picks the winning configuration once each candidate has been scored.

Reference: Satopaa et al. (2011), Finding a "Kneedle" in a Haystack; Rousseeuw (1987).

What to Observe

- **Two separate, composable concerns:** evaluation customization and hyperparameter tuning — you can change one without the other.
- **Same experiment shape:** both examples keep configure → fit → cluster → evaluate intact; tuning changes how the configuration is chosen, not how the experiment is read.



Acknowledgements



Data Analytics Lab

dal.global@grupo.cefet-rj.br | <https://eic.cefet-rj.br/dal>